

Paper Code



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Faculty of Engineering, Mathematics and Science

School of Computer Science & Statistics

MSc in Computer Science Hilary Term 2026 Year 1

Security of Networks and Distributed Systems

CS7NS5/CSU44032

dd MMM 2026

Location

Time (2 hours)

Dr. Stephen Farrell

Instructions to Candidates:

Attempt **three** questions. All questions carry equal marks (40 each).

You may not start this examination until you are instructed to do so by the invigilator.

Materials Permitted for this examination:

N/A

Question 1

An email newsletter distribution company (let's call them "distro") send bulk emails on behalf of their customers, generally to lists of email addresses supplied by those customers. As the customers manage the lists of email addresses, the customers are the GDPR data controllers for those lists. Newsletter content can be supplied by the customer or generated on their behalf as a (paid-for) editorial service from "distro." Since LLM tooling has become widely available, more customers are generating their own newsletter content using such tools. Even when a customer prepares newsletter content, "distro" still currently need to modify that content, so that e.g. plain content URLs are turned into tracking URLs that feed into the telemetry "distro" make available to their customers.

"Distro" however live or die based on the deliverability of their email, which is non-trivial when sending bulk email. Deliverability is also mostly out of the sender's control and is ultimately controlled by email recipients classifying the emails as spam or ham, and by mailbox providers such as Gmail or Microsoft.

Given the increasing use of LLM tools by "distro" customers, the company is becoming more concerned about how that might create risks for their business and so they engage you to carry out a risk analysis for them, to consider the impact of LLMs and other technical ecosystem developments and how those might impinge on the "distro" business model.

(a) Describe the process you would use to carry out this task (note: this part of the question asks about process, not the specific risks associated with the new environment).

[15 marks]

(b) Describe three significant technical risks affecting the "distro" business model, including their potential impact and likelihood of occurrence, and outline the related mitigations you would recommend that "distro" ought to consider.

[15 marks]

(c) Any genuine bulk emailed newsletter will include an "unsubscribe" link allowing the email recipient to opt-out of future versions of the newsletter. Good distribution companies (like "distro") will also offer the option of opting-out of all future newsletters and other marketing communications, across all "distro" customers. Who is the GDPR data controller for the list of email addresses that have opted out of all communication? Say why that is the case, and describe two consequences that flow from that for "distro" and their customers.

[10 marks]

Question 2

(a) For any real Internet key exchange protocol (e.g. TLS, IPsec, or Kerberos) describe the key management and application data protection aspects of the protocol in detail. Describe how a library developer, for example the OpenSSL project for the TLS protocol, might test their implementation.

Your answer may describe any widely used version of your chosen protocol, but for TLS, you must explicitly state which version you are describing.

[25 marks]

(b) Describe how cryptographic algorithms intended to withstand attacks using a Cryptographically Relevant Quantum Compute (CRQC) could affect deployments of your chosen protocol. Aside from what “could” happen – what “should” be done in the next 3-5 years? (And why.)

[10 marks]

(c) Describe one realistic way in which some load-balancing technique can be used as a way to improve the scalability of a deployment of your chosen protocol.

[5 marks]

Question 3

You are tasked to design a new web-based system for handling legal documents for the Irish courts system. The system must allow lawyers and judges involved in cases to file and access documents as cases progress. The public should also be able to access filings and evaluate their authenticity (as filed documents) without authenticating. Typical filings might include evidence statements, pleadings, and decisions. Some non-lawyers will also be able to login to the system, for example, legal secretaries and court staff who can file documents on behalf of relevant lawyers or judges. It is critical that only appropriately authorised users can modify the state of the system, and that all filings retrieved from the system can be demonstrated to have been validly filed. You can assume there are about 5,000 users who can login to the system and that it processes some hundreds of thousands of filings annually. For parts (a) and (b) of your answer you may assume that all filings are public information and immutable.

Notes: 1) If you are not familiar with the Irish legal system, it is entirely fine to base your answer on some other legal system with which you are more familiar. 2) Don't get bogged down in too much detail – good descriptions of the requirements and most important mechanisms proposed are what's important in your answer. 3) Pay attention to the marks for each section of your answer.

Lastly, you might consider how this system could have similarities with how Certificate Transparency is used as part of the WebPKI.

(a) Outline a design for the overall system (include role descriptions and diagrams as appropriate), and state the security and privacy requirements the overall system should meet. Note that this part of your answer should only discuss security and privacy requirements and not describe specific mechanisms to meet those requirements.

[20 marks]

(b) Describe, in detail, the most important security and privacy mechanisms you propose for the system to meet requirements from part (a).

[15 marks]

(c) Some legal cases involve information that is not public, for example, cases involving children or where the court orders that information not be published until after the case has been decided. How could you extend your system to support such non-public information that needs to be verifiable and available to some users, but that won't become public until later, if ever?

[5 marks]

Question 4

(a) Describe the privacy issues that arise with the Domain Name System (DNS), and usable mitigations for those, including those that might affect registrants, as well as those arising from use of the DNS for querying resource records. Where relevant, include discussion of the pros and cons of various deployment options, such as the use of public recursive resolvers.

[20 marks]

(b) Describe the ways in which PKIs, and in particular the Web PKI, interact with the DNS and how those impact on the security and privacy of the CAs, relying-parties and subscribers involved.

[15 marks]

(c) What is the most practical and impactful way in which you could abuse the DNS for fun or profit (or to be a nuisance)? How would you execute your evil plan?

[5 marks]